A Note on Binary Arrays with a Certain Window Property

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***Abstract*—For some $rs = 2^{mn}$, there exist periodic $r \times s$ binary arrays such that all $m \times n$ binary matrices will appear in it. As two-dimensional versions of de Bruijn sequences, these kinds of arrays have possible applications in cryptography and the design of mask configurations for spectrometers.**

Reed and Stewart [7], Spann [8], and Gordon [2] have studied the two-dimensional arrays of so-called perfect maps. This has led to research on various types of window properties for binary array (see [5], [6], and [9]). This correspondence presents some methods for the construction of constructing arrays having a certain window property. These arrays can be regarded as two-dimensional versions of de Bruijn sequences or maximal-length shift register sequences [1]. The analogy to the one-dimensional case suggests an application in cryptography for encrypting two-dimensional arrays, e.g., images. The arrays can also be used to design mask configurations for spectrometers [3].

A periodic $r \times s$ binary array is an infinite matrix $A = (a_{ij})$ with $i, j = 0, 1, 2, \cdots$, where a_{ij} is either zero or one and $a_{ij} = a_{i+r,j} = a_{i,j+s}$ for all i, j. In particular if, for $0 \le i < r$ and $0 \le j < s$,

$$\begin{bmatrix} a_{ij} & a_{i,j+1} & \cdots & a_{i,j+n-1} \\ a_{i+1,j} & a_{i+1,j+1} & \cdots & a_{i+1,j+n-1} \\ \cdots & \cdots & & \cdots \\ a_{i+m-1,j} & a_{i+m-1,j+1} & \cdots & a_{i+m-1,j+n-1} \end{bmatrix}$$

represents each $m \times n$ binary matrix exactly once, then A has $m \times n$ window property. Our window property is slightly different from that defined in [5] in that we include the all-zero matrix. Obviously, if there is a periodic $r \times s$ binary array with $m \times n$ window property, then $rs = 2^{mn}$. Hence, such an array is a perfect map for $m \times n$ submap with maximal size.

Let $M(p, q)$ denote the set of all q-ary sequences $\boldsymbol{u}$ of period q^p such that, for $0 \le k < q^p$, $(u_k, u_{k+1}, \cdots, u_{k+p-1})$ represents all q-ary p-tuples exactly once. Also, define the map of linear shift T on the set of all sequences such that $T^k \boldsymbol{u} = \boldsymbol{v}$, where $v_i = u_{i+k}$ for all i. Let $\boldsymbol{u}^t$ denote the transpose of $\boldsymbol{u}$.

Lemma 1: Let $\boldsymbol{v} \in M(n-1, 2^m)$ for $n \ge 3$, and define $\boldsymbol{w}$ by $w_0 = 0$ and $w_i = v_0 + v_1 + \cdots + v_{i-1}$ modulo 2^m for $i > 0$. Then, $\boldsymbol{w}$ is of period $2^{m(n-1)}$.

Proof:

$$v_0 + v_1 + \cdots + v_{2^{m(n-1)}-1} \equiv w_{2^{m(n-1)}} \bmod 2^m.$$

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

But

$$v_0 + v_1 + \cdots + v_{2^{m(n-1)}-1} \equiv 0 \bmod 2^m$$

provided $n \ge 3$. Hence, $\boldsymbol{w}$ is of period $2^{m(n-1)}$.

Theorem 1: Let $\boldsymbol{u} \in M(m, 2)$, $\boldsymbol{v} \in M(n-1, 2^m)$ for $n \ge 3$, and let $\boldsymbol{w}$ be defined as in Lemma 1. Let $N = 2^{m(n-1)}$. Then,

$$A = [T^{w_0}\boldsymbol{u}^t, T^{w_1}\boldsymbol{u}^t, \cdots, T^{w_{N-1}}\boldsymbol{u}^t, T^{w_N}\boldsymbol{u}^t, \cdots]$$

is a periodic $2^m \times 2^{m(n-1)}$ binary array with $m \times n$ window property.

Proof: Since $\boldsymbol{u}$ has period 2^m and $\boldsymbol{w}$ has period $2^{m(n-1)}$, it follows that $a_{ij} = a_{i+2^m,j} = a_{i,j+2^{m(n-1)}}$; thus, A is a periodic $2^m \times 2^{m(n-1)}$ binary array. If B is a $m \times n$ binary matrix, then every column of B appears somewhere in $\boldsymbol{u}$, and it is possible to find integers $r_1, r_2, \cdots, r_{n-1}$, $0 \le r_i \le 2^m - 1$ such that B appears in the semi-infinite matrix

$$[\boldsymbol{u}^t, T^{r_1}\boldsymbol{u}^t, T^{r_1+r_2}\boldsymbol{u}^t, T^{r_1+r_2+r_3}\boldsymbol{u}^t, \cdots, T^{r_1+r_2+\cdots+r_{n-1}}\boldsymbol{u}^t].$$

Since $\boldsymbol{v} \in M(n-1, 2^m)$, $(r_1, r_2, \cdots, r_{n-1}) = (v_k, v_{k+1}, \cdots, v_{k+n-2})$ for some k, which implies that B appears in the semi-infinite matrix

$$[T^{w_k}\boldsymbol{u}^t, T^{w_{k+1}}\boldsymbol{u}^t, \cdots, T^{w_{k+n-1}}\boldsymbol{u}^t],$$

that is, B appears in A. Q.E.D.

We can construct sequences belonging to $M(m, 2)$ and $M(n-1, 2^m)$ from de Bruijn sequences [1]. As an example of the results obtainable from the theorem, we have (0 0 1 1) in $M(2, 2)$ and (0 0 1 1 2 2 3 3 0 2 0 3 1 3 2 1) in $M(2, 4)$. Hence, the first 4×16 entries of a periodic array with 2×3 window property are as shown below:

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

Theorem 2: Let $[\boldsymbol{u}^t, \boldsymbol{v}^t, \boldsymbol{u}^t, \boldsymbol{v}^t, \cdots]$ be a periodic $2^{m-1} \times 2$ binary array with $m \times 1$ window property, and define

$$A_i = [T^{r_i}\boldsymbol{u}^t, T^{r_i+i}\boldsymbol{u}^t, T^{r_i+2i}\boldsymbol{v}^t, T^{r_i+3i}\boldsymbol{v}^t]$$

for $r_i = 4\{1 + 2 + \cdots + (i-1)\}$. Then,

$$[A_0, A_1, A_2, \cdots]$$

is a periodic $2^{m-1} \times 2^{m+1}$ binary array with $m \times 2$ window property.

The proof of Theorem 2 is similar to that of Theorem 1 and is omitted. As an example, the periodic array whose first 4×2 entries are as shown below has the 3×1 window property:

$$\begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Hence, the first 4×16 entries of a periodic array with 3×2 window property are as shown below:

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

With Theorem 2, it is natural to ask whether $2^{m-1} \times 2$ arrays exist with $m \times 1$ window property. The answer is positive if $m \ge 3$.

A de Bruijn diagram or Good diagram (see [1] and [4]) is the directed graph $G_p = (V_p, E_p)$, where V_p is the set of all binary p-tuples, and $(\boldsymbol{a}, \boldsymbol{b})$ is in E_p if $a_{i+1} = b_i$ for $i = 1, 2, \cdots, p-1$. A path $\{(\boldsymbol{a}^{(1)}, \boldsymbol{a}^{(2)}), (\boldsymbol{a}^{(2)}, \boldsymbol{a}^{(3)}), \cdots, (\boldsymbol{a}^{(k-1)}, \boldsymbol{a}^{(k)})\}$ in G_p will then correspond to a finite sequence $(a_p^{(1)}, a_p^{(2)}, \cdots, a_p^{(k)})$. Thus, the existence of a $2^{m-1} \times 2$ array with $m \times 1$ window property is equivalent to the existence of two disjoint closed paths (or cycles) in G_m of equal length that together pass through all vertices in V_m exactly once.

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Proposition 1: If $m \geq 3$, then there exists a $2^{m-1} \times 2$ binary array with $m \times 1$ window property.

Proof: Let D: $G_m \rightarrow G_{m-1}$ such that $D\boldsymbol{a} = \boldsymbol{b}$ if $b_i \equiv a_i + a_{i+1} \bmod 2$ for $i = 1, 2, \cdots, m-1$. Let C be a complete cycle in G_{m-1}, that is, a closed path passing through all vertices in V_{m-1} exactly once. The existence of C is equivalent to the existence of a de Bruijn sequence of span $m-1$ and these are known to exist. If $m - 1 \geq 2$, then by Corollary 1 of [4], the preimage of C under D consists of two disjoint cycles of equal length and hence our proposition follows. Q.E.D.

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Coin Flipping by Telephone

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***Abstract*—Two parties, who do not trust each other, wish to agree on the result of a coin toss while they are conversing by telephone. A new protocol which solves this problem is proposed and compared with previous protocols.**

I. Introduction

Blum [4] considered the problem of two parties who wish to agree on the result of a coin toss while they are conversing by telephone. These two parties, A and B, do not trust each other and do not have the services of an intermediary. He suggested a protocol that solves the problem using Rabin's method of "oblivious transfer" [3]. The method relies on the difficulty of computing square roots modulo N, where $N = pq$, a product of two unknown primes. If p and q were known, computing square roots would be easy, thus squaring modulo N is a trapdoor one-way function as defined in [1]. Shamir, Rivest, and Adleman [2] described how to play "mental poker," and their scheme can be used for coin flipping. They require trapdoor commutative one-way functions.

Our solution to the problem assumes the existence of two functions, f_h and f_t ("h" stands for "head," and "t" for "tail"), with the following properties:

a) Given y, it is computationally impossible to determine an x such that $y = f_h(x)$ or $y = f_t(x)$, (i.e., f_h and f_t are one-way functions).
b) It is computationally impossible to find x and x' such that $f_h(x) = f_t(x')$.
c) Both f_h and f_t have the same range, and each element in this range has the same number of inverse images under both f_h and f_t.

Once A and B have such a pair of functions, the coin-flipping protocol is straightforward:

1) A selects some element x and sends either $y = f_h(x)$ or $y = f_t(x)$ to B.
2) B guesses which function, f_h or f_t, was used and sends his guess to A.
3) A transmits x to B. B computes $f_h(x)$ and $f_t(x)$ to confirm that A is playing fairly and to determine whether he won the toss by guessing correctly in step 2).

Our method does not require that the one-way functions be commutative or have the trapdoor property. In addition, the set of properties we listed earlier can be relaxed considerably, especially property a). We listed them as we did mainly to simplify the subsequent discussion. For example, the one-way functions may be implemented by using the data encryption standard [6]. We could choose $f_h(x) = \mathrm{DES}(x, P_h)$, the encryption of the fixed plaintext P_h using the key x, and similarly we could choose for $f_t(x)$. Although the ranges of these two functions are not necessarily the same, they would probably serve well in our protocol.

The requirements of our method are weak enough that a large number of one-way functions can satisfy them. At present we can give only one example where the protocol can be proved secure. To do so, we assume that discrete exponentiation is a one-way function.

II. Example

Exponentiation modulo a prime p is used as a one-way function in the following protocol:

1) A and B mutually choose a prime number p in such a way that the factorization of $p - 1$ is known.
2) B selects two primitive elements h and t in $\mathrm{GF}(p)$.
3) A chooses an integer x relatively prime to $p - 1$ and sends B an integer y which is either $h^x \pmod p$ or $t^x \pmod p$.
4) B guesses h or t and sends his guess to A.
5) A reveals x to B. B computes $h^x \pmod p$ and $t^x \pmod p$ to confirm that A is playing fairly, and to determine whether he won the toss by guessing correctly on step 4).

In step 1 of the protocol we require that p be chosen in such a way that the factorization of $p - 1$ is available to both parties. For example, A and B could pick p by starting from a product of known primes, then adding 1 and checking for primality. Knowing the factorization of $p - 1$ helps assess the difficulty of computing logarithms in $\mathrm{GF}(p)$. If $p - 1$ has only small prime factors, computing logarithms is easy using the Pohlig–Hellman algorithm [5]. Hence, when p is chosen, B will insist that this will not happen. We also use the factorization of $p - 1$ to check the primitivity of h and t, which is desirable for reasons mentioned below. No efficient algorithm for checking primitivity is known unless the factorization of $p - 1$ is specified.

To show that this protocol is reliable, let us examine whether or not the parties can cheat. A can cheat if he knows two integers x and x' such that $h^x = t^{x'} \pmod p$. To know x' when x is given, A has to compute $\log_t h^x \pmod{p-1}$. This would have

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